

04.05 Solving Radical Equations

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
A variable is under one radical.	Isolate the radical, square both sides, solve, and check.
Two radicals remain.	Isolate one radical at a time and repeat the process.
A candidate makes a radicand negative or fails the original equation.	Reject it as invalid or extraneous.

A. Solve one-radical equations

1.

$$\sqrt{x+1} = x-1$$

2.

$$\sqrt{x+4} = x-2$$

3.

$$\sqrt{2x+3} = x$$

4.

$$\sqrt{x-3} = 5$$

5.

$$\sqrt{3x - 2} = 4$$

6.

$$\sqrt{x + 6} = x$$

7.

$$\sqrt{2x - 1} = x - 1$$

8.

$$\sqrt{x + 9} = x + 1$$

9.

$$\sqrt{4x + 1} = x + 1$$

10.

$$\sqrt{x - 2} + 1 = 4$$

11.

$$\sqrt{2x + 5} = 3$$

12.

$$\sqrt{x + 2} = 2x - 1$$

13.

$$\sqrt{3x + 1} = x + 1$$

14.

$$\sqrt{x - 5} = x - 7$$

15.

$$\sqrt{2x + 8} = x$$

16.

$$\sqrt{x + 7} = x - 1$$

17.

$$2\sqrt{x+1} = 6$$

18.

$$\sqrt{5x-4} = x$$

19.

$$\sqrt{x+12} = x$$

20.

$$\sqrt{2x+1} = x+1$$

B. Solve equations with two radicals

1.

$$\sqrt{x+5} - \sqrt{x} = 1$$

2.

$$\sqrt{x+8} = \sqrt{x} + 2$$

3.

$$\sqrt{x+9} - \sqrt{x} = 3$$

4.

$$\sqrt{x+4} + \sqrt{x} = 4$$

5.

$$\sqrt{2x+1} = \sqrt{x+4}$$

6.

$$\sqrt{x+6} = \sqrt{x} + 1$$

7.

$$\sqrt{x+10} - \sqrt{x+1} = 1$$

8.

$$\sqrt{x + 12} = \sqrt{x} + 2$$

9.

$$\sqrt{2x + 3} - \sqrt{x} = 1$$

10.

$$\sqrt{x + 15} = \sqrt{x} + 3$$

11.

$$\sqrt{x + 16} - \sqrt{x} = 2$$

12.

$$\sqrt{x + 20} = \sqrt{x} + 4$$

13.

$$\sqrt{3x + 1} = \sqrt{x + 5}$$

14.

$$\sqrt{x+7} - \sqrt{x+3} = 1$$

15.

$$\sqrt{2x+8} = \sqrt{x} + 2$$

16.

$$\sqrt{x+25} - \sqrt{x} = 5$$

17.

$$\sqrt{x+18} = \sqrt{x} + 3$$

18.

$$\sqrt{2x+12} = \sqrt{x} + 2$$

19.

$$\sqrt{x+24} - \sqrt{x} = 4$$

20.

$$\sqrt{3x + 4} = \sqrt{x + 8}$$

C. Check for extraneous solutions

1.

$$\sqrt{x + 3} = x - 3$$

2.

$$\sqrt{x + 5} = x - 1$$

3.

$$\sqrt{2x + 4} = x$$

4.

$$\sqrt{x - 1} = 3 - x$$

5.

$$\sqrt{4x + 1} = x - 1$$

6.

$$\sqrt{x + 8} = 2 - x$$

7.

$$\sqrt{3x - 2} = x - 2$$

8.

$$\sqrt{x + 6} = 4 - x$$

9.

$$\sqrt{2x + 7} = x + 1$$

10.

$$\sqrt{x - 4} = x - 6$$

11.

$$\sqrt{x + 10} = x - 2$$

12.

$$\sqrt{5x - 4} = 3 - x$$

13.

$$\sqrt{x + 2} = x - 4$$

14.

$$\sqrt{2x + 1} = x - 3$$

15.

$$\sqrt{x + 12} = 5 - x$$

16.

$$\sqrt{3x + 5} = x - 1$$

17.

$$\sqrt{x-2} = x-5$$

18.

$$\sqrt{4x+9} = x+1$$

19.

$$\sqrt{x+14} = x-3$$

20.

$$\sqrt{2x+5} = 4-x$$